

15-122: Principles of Imperative Computation

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<https://cs.cmu.edu/~15122>

Overview

- Goals of this course
- Interactions
 - Lectures, precepts, office hours
- Practice
 - Homework, exams, activities
- The course begins ...

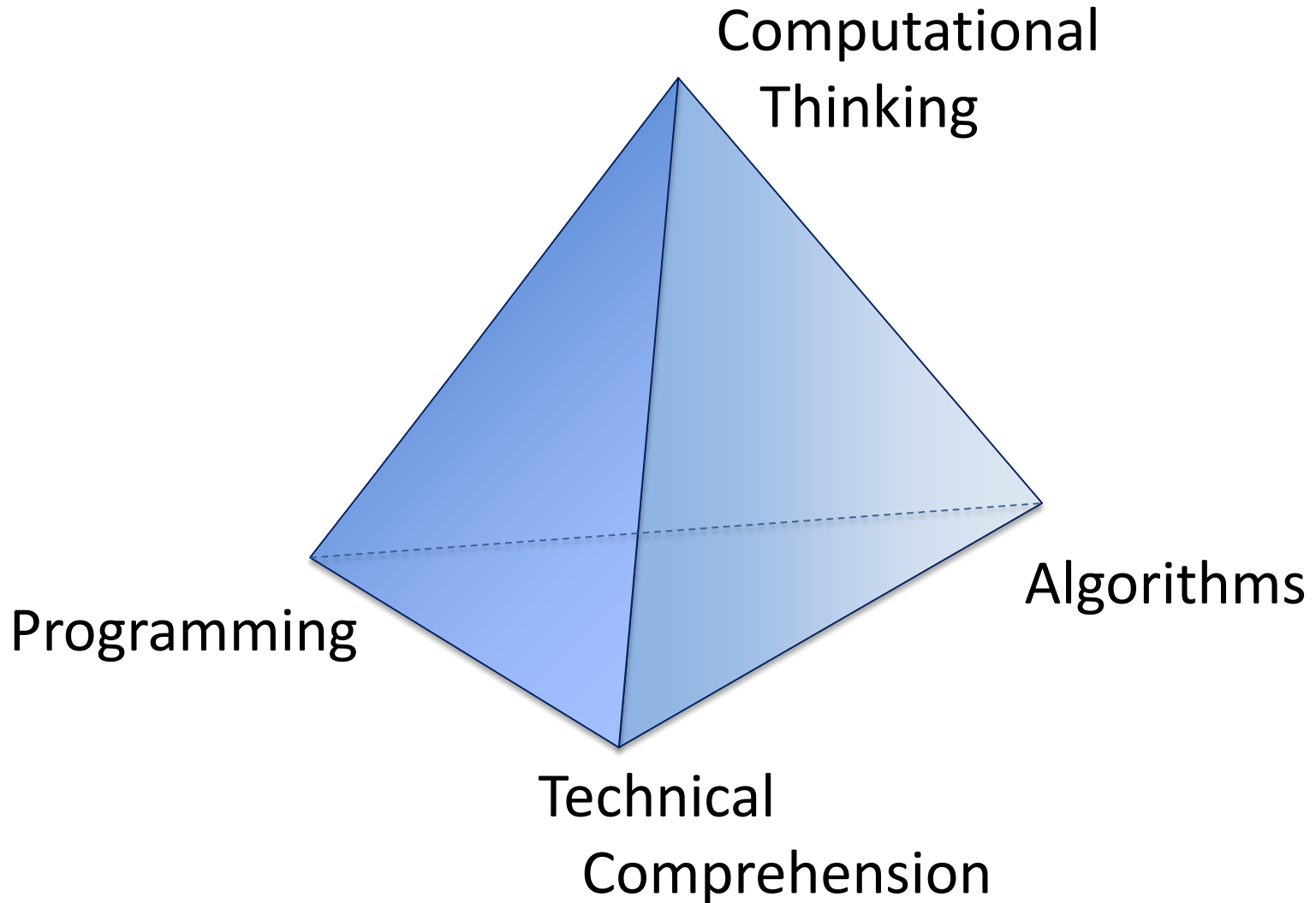
What you will get out of 15-122

- **Confidence** to write small programs correctly
 - up to a couple thousand lines of code
- Familiarity with lots data structures
 - and algorithms too
- (Some) experience with C
- Systematic approach to solving problems
- Good time management

Should you be taking this course?

- Yes!
 - it is a core requirement for your major
 - Computer Science is your passion
 - you developed applications independently
- No.
 - you haven't yet mastered 15-112's content

Goals



Programming Skills

- Transforming algorithmic ideas into code
 - Code that works the first time around
 - *Deliberate programming*
 - ... well, *nearly* the first time around
 - Testing
- Imperative programming in C and C0
- Basic Unix skills

Algorithmic Knowhow

- Shared vocabulary
 - safety, correctness
 - worst case/average case/amortized complexity
 - order, randomness
- Lots of fundamental data structures
- Lots of algorithms

(Psst... this is often what tech interviews test on!)

Computational Thinking

- From ***programmer*** to **computer scientist**
 - Systematic approach to solving a problem
 - Finding solutions that are *correct*
 - Finding solutions that are *efficient*

Technical Comprehension

- Reading technical specifications
 - Problem statements will get longer
 - Dots will be further apart
 - + You will become **more confident**
 - + You will try more things on your own

The Big Picture

- Pre- or co-requisites
 - either 15-151 (Math Foundations for CS)
 - or 21-127 (Concepts of Mathematics)
- Counterpart
 - 15-150 (Principles of Functional Programming)
- Pre-requisite for
 - **15-213 (Introduction to Computer Systems)**
 - 15-210 (Parallel and Sequential Data Structures and Algorithms)
 - 17-214 (Principles of Software Construction)

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Lectures

- Tuesdays and Thursdays
- Please be here, please be active
 - ask and answer questions, pay attention
 - lecture notes and slides for review
- Laptops for note-taking only
 - In the back unless you can't see the board
 - Too distracting for other students
 - No surfing, email, games, homework, ...

Friday Precepts

- Practice the week's topics in small groups
 - with TA support
- Choose the problems you want to work on
 - Tip: *pick the ones you are **least confident** about!*
 - Use the rest later for further practice
- Graded based on effort
 - Group gets grade based on members' understanding

Attend the precept you're registered for

Getting-started Help

- 15-122 setup — set up the C0 tools with Andrew Linux
 - **Either** setup lab
 - See welcome email
 - **or** laptop setup
 - Tonight, drop in between 7pm and 8:20pm in **GHC 4401**
- Linux workshop — Learn useful Linux commands
 - Wednesday 5 to 7pm in **GHC 4401**

Online Resources

- Course home page <http://cs.cmu.edu/~15122>
 - Schedule, calendar, contact info...
 - Lecture notes, slides
 - Assignments, precepts, solutions
 - Links to all resources
- C0 home page <https://c0.cs.cmu.edu>
 - Tutorial, reference, examples

Online communication

- **Ed** for announcements, questions, and communication with course staff
 - Get help, help each other!
 - Make your posts **public**
 - unless discussing solutions
- **Autolab** and **Gradescope** for homework
- Grades from [course home page](#)
- SSH to shared machines for assignments
 - from your laptop or a cluster computer

Help through the Semester

- **Office hours**

Starts Friday

- Calendar on [course web page](#)

- **Bootcamps**

First one
this Saturday
11am-2pm

- **Student Academic Success Center**

- [Supplemental Instruction](#)

- [Peer Tutoring:](#)

- 1-on-1
 - drop-in

Details soon

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Practice

- Daily in-class activities and weekly precepts
 - In-class activities in lectures
- Weekly Practice Problems
 - due **Thursdays by 6pm on Gradescope**
 - Download assignment from course web page
 - No late days
 - ∞ submissions
- Weekly Programming Assignments
 - due **Sundays by 9pm on Autolab**
 - Download assignments and starter code from Autolab
 - 3 late days, at most 1 per homework
 - Extensions only for emergencies
- Weekly checkins
 - **Tuesdays 7:00-7:35pm or 7:45-8:20pm** in GHC 4401, DH 2210 or DH 2315
- Final exam (TBA)



PP 0, due
this Thursday

Grades

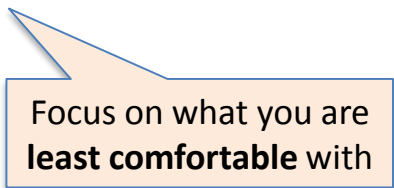
Points

- 720 pt – checkins (CH)
 - 780pt available (bucket system)
- 720 pt – Homework (HW)
 - 120 pt - Practice Problems
 - 600 pt - Programming Assignments
- 30 pt - In-class activities and precepts (AC)
 - 39pt available (bucket system)
- 250 pt - Final exam (FI)

Course performance

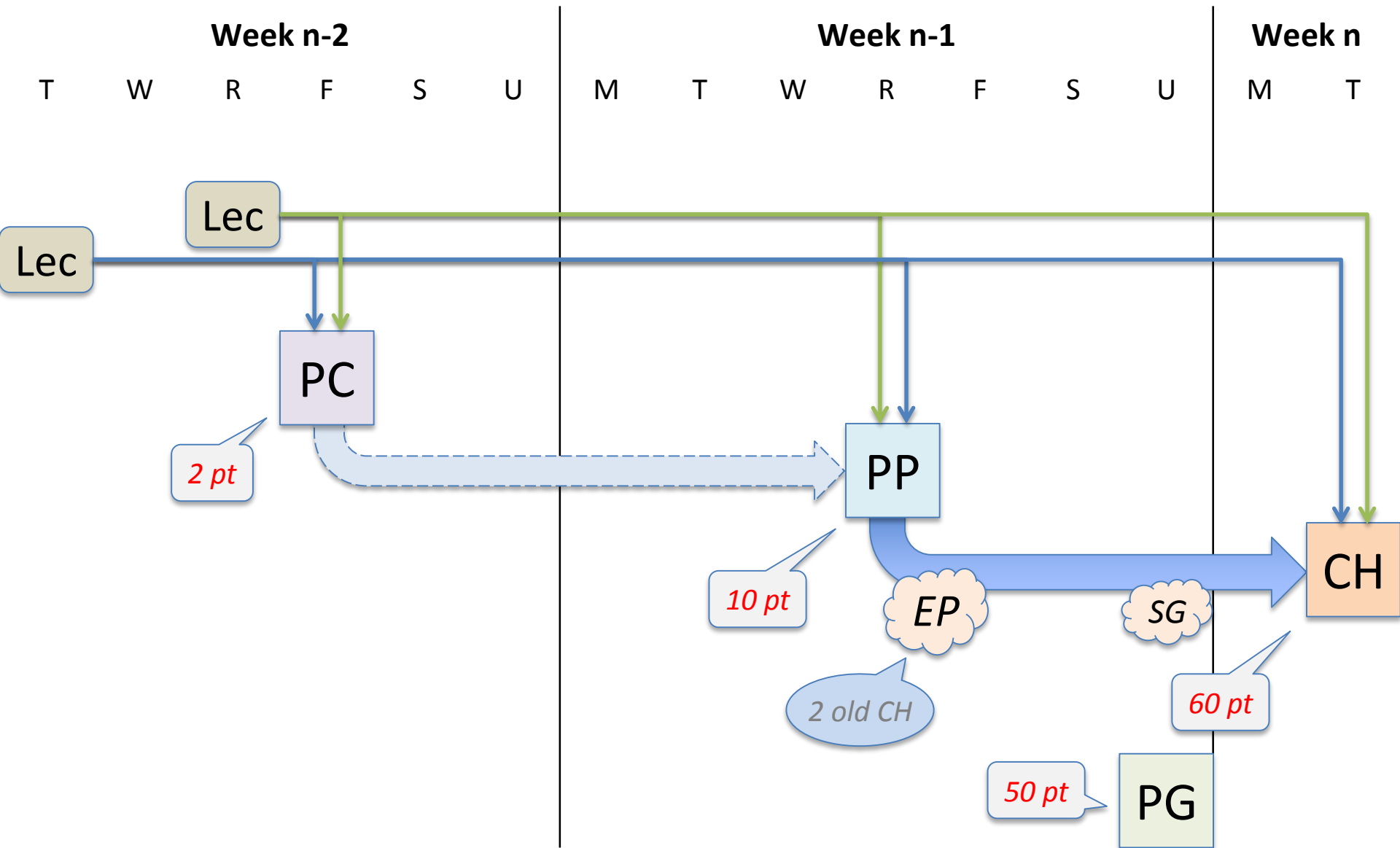
$$\min(\text{CH}, \text{HW}) + \text{AC} + \text{FI}$$

- max 1,000
- divide by 10 for %



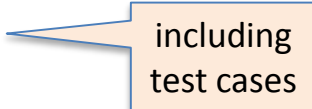
Focus on what you are
least comfortable with

Workflow



Academic Integrity

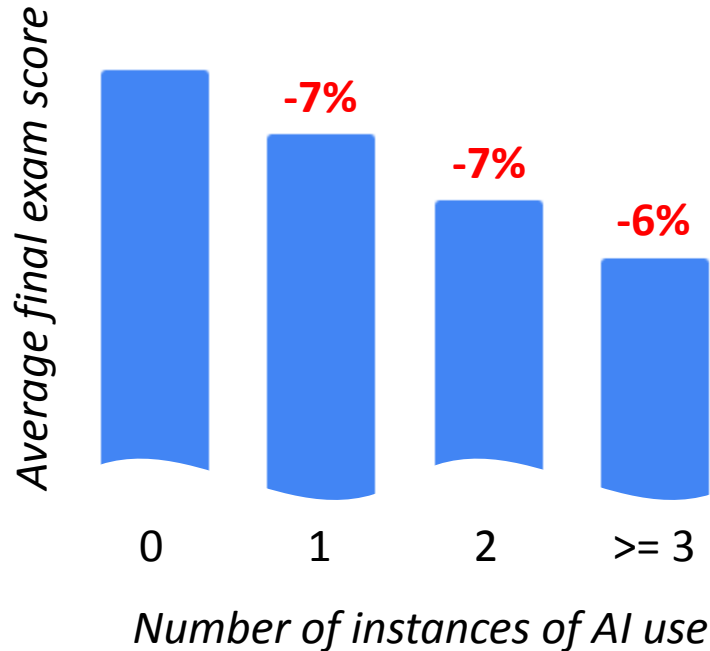
- Practice Problems:
 - collaborative work
 - individual submissions
- Programming hw *must be your own*
 - No collaboration
 - No outside help (friends, web, AI)
- Made a mistake? Email us!



including
test cases

You are here to learn, not to get a grade

Using AI in 15-122



- Most people didn't use AI
- The majority of ≥ 3 didn't pass 15-122
- Doesn't count drops

- AI is useful for tasks you've done a million times
 - Not for learning it the first time around

AI Assistance

- Not allowed

- not for writing code
 - including test cases
- not for checking answers
- not for brainstorming
- never for homework



Use
Ed and OH
instead

How to do Well in this Course

- Do not stress over grades
- Participate
- Try things out
- Manage your time wisely
 - *Don't use late days in 1st half of course*
- Study the material steadily
- Start homework early
- Get all the help you need
 - *ask for help, tell us when you're having trouble*
- Make time for fun

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